

Material Request Management System

AWS Agentic AI | 150 Employees | Multi-tier Approvals | Real-time Dashboards

Amazon Bedrock

AWS Cognito

API Gateway

Lambda

DynamoDB

S3

Textract

CloudWatch

SNS

AppSync

LAYER 1 — EMPLOYEE



Secure Login

JWT · 150 unique users
Cognito User Pool

Material Request Form

Auto-fill: Name, EmpID, Date
Project · Category · Qty

Document Upload

Supporting docs (PDF/Image)
S3 pre-signed URL

Submit Request

DynamoDB record created
SNS notification triggered

Submit form + docs

LAYER 2 — AWS AGENTIC AI (Amazon Bedrock)



Bedrock AI Agent

Amazon Bedrock (Claude)
Orchestrates validation

Document Validation

Textract extracts doc data
Checks vs form fields

Smart Classification

Budget category inference
Anomaly detection

Intelligent Routing

Routes to correct manager
Priority scoring

Validated & routed

LAYER 3 — MANAGER



Manager Login

Role-based access (RBAC)
Cognito Manager Group

Approval Inbox

Pending · In Review · Decided
Real-time via AppSync

Review & Compare

Form vs Doc side-by-side
AI match confidence score

Approve / Reject

Approve → notify & update
Reject → mandatory comment

Approved / escalated

LAYER 4 — DIRECTOR / EXECUTIVE



Executive Login

Director / C-Suite access
Org-wide visibility

Org Dashboard

All projects & depts
Budget consumed vs allocated

Project Analytics

Per-project approvals
Monthly trend & top requesters

AI Insights Panel

Anomaly flags & forecast
Bedrock-generated summaries

LAYER 5 — AWS CLOUD INFRASTRUCTURE



Amazon Cognito

Auth · User pools · RBAC
MFA support

API Gateway + Lambda

RESTful APIs · Serverless
Event-driven microservices

DynamoDB + S3

Form records · Doc storage
Encryption at rest

CloudWatch + SNS

Logs · Alerts · Email/SMS
Audit via CloudTrail

Decision Logic & AWS Agentic AI Rules

All decision points driven by Amazon Bedrock Agent

01 DECISION 01 | Document Matching

YES / PASS

AI confidence $\geq$ 80% -> Auto-pass to manager inbox

NO / FAIL

Below threshold -> Flag for manual review by manager

02 DECISION 02 | Manager Decision

YES / PASS

Approve -> Notify employee + update project budget

NO / FAIL

Reject -> Mandatory comment required -> Notify employee

03 DECISION 03 | Budget Threshold Check

YES / PASS

Within project limit -> Standard approval flow continues

NO / FAIL

Exceeds threshold -> Auto-escalate to Director level

MANDATORY REJECTION COMMENT - UX RULE

When a manager selects Reject, the comment field becomes required. The form cannot be submitted without a reason. This ensures accountability and gives employees actionable feedback. Comment is stored in DynamoDB, visible to employee and all escalated levels.

Bedrock AI Agent analyses rejection patterns over time for process improvement insights.

AMAZON BEDROCK AGENT - CAPABILITIES IN THIS SYSTEM

- Extract & validate fields from uploaded documents (Textract)
- Cross-check document data with submitted form values
- Score confidence of document-form match (0-100%)
- Detect duplicate requests from same employee
- Route to correct manager based on org hierarchy
- Flag budget anomalies vs historical spend patterns
- Auto-categorise material type for budget allocation
- Generate natural-language summaries for director dashboard

MANAGER DASHBOARD

Project-level budget tracking | Approval & rejection analytics | Team activity | Real-time updates via AppSync

KEY METRICS (This Month)



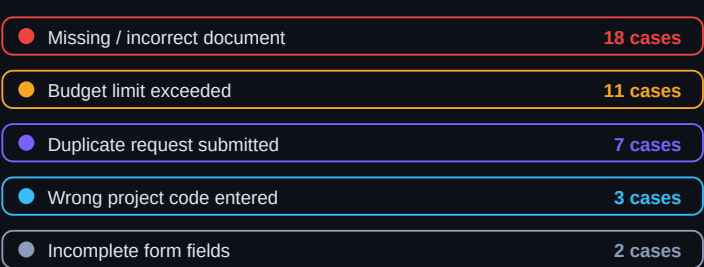
PROJECT-WISE BUDGET TRACKER

Project Name	Allocated	Consumed	Remaining	Requests	Budget Used	Status
Office Renovation Phase 2	AED 80K	AED 61K	AED 19K	38	76%	Active
IT Infrastructure Upgrade	AED 50K	AED 47K	AED 3K	22	94%	Near Limit
Warehouse Fit-Out	AED 120K	AED 58K	AED 62K	51	48%	Active
Marketing Campaign Q2	AED 30K	AED 16K	AED 14K	14	53%	Active
Vehicle Fleet Maintenance	AED 25K	AED 25K	AED 0	11	100%	Closed

APPROVAL BREAKDOWN BY PROJECT



TOP REJECTION REASONS



RECENT REQUESTS INBOX (Pending Manager Action)

Employee Name	Project	Material	Qty / Cost	Doc Match	Submitted	Action
Ahmed Al Mansoori	Office Renovation	Steel brackets	50 / AED 1,200	95%	10 Jun	PENDING
Priya Krishnan	IT Infrastructure	SFP Transceivers	12 / AED 3,600	88%	10 Jun	PENDING
Khalid Rashid	Warehouse Fit-Out	Safety signage	30 / AED 450	72%	09 Jun	REVIEW
Sara Mohamed	Marketing Q2	Printed banners	100 / AED 800	91%	09 Jun	PENDING
Ravi Shankar	IT Infrastructure	Patch cables	200 / AED 600	99%	08 Jun	PENDING

BEDROCK AI INSIGHTS FOR MANAGER

- IT Infrastructure budget is 94% consumed - 2 pending requests may exceed limit. Consider escalating to Director.
- Document rejection rate rose 12% vs last month. Main cause: missing supplier invoices in upload.
- Ahmed Al Mansoori has 3 pending requests this week - possible duplicate detected in materials list.

DIRECTOR / EXECUTIVE DASHBOARD

Organisation-wide visibility | All projects & departments | AI-generated summaries via Amazon Bedrock

ORGANISATION-WIDE KPIs



12

Active Projects



1,847

Total Requests (YTD)



74%

Approval Rate



AED 2.4M

Total Budget Allocated



AED 1.1M

Budget Consumed (YTD)



7 Active

AI Anomalies Flagged

DEPARTMENT-WISE BUDGET UTILISATION



ALL PROJECTS OVERVIEW

Project	Manager	Budget	Consumed	Requests	Approved	Rejected	Status
Office Renovation Ph2	Ibrahim K.	AED 80K	AED 61K	38	29	9	Active
IT Infrastructure	Sara M.	AED 250K	AED 235K	22	17	5	At Risk
Warehouse Fit-Out	Rajan P.	AED 500K	AED 275K	51	39	12	Active
Marketing Campaign Q2	Leila A.	AED 200K	AED 126K	14	11	3	Active
Fleet Maintenance	Khalid R.	AED 25K	AED 25K	11	8	3	Closed
ERP System Rollout	Priya K.	AED 350K	AED 89K	9	7	2	Active
Office Renovation Ph3	Ibrahim K.	AED 120K	AED 12K	5	4	1	New

AWS SERVICES POWERING DASHBOARDS

- QuickSight:** Charts, KPIs, drill-down for directors
- DynamoDB Streams:** Real-time dashboard data refresh
- AppSync WebSocket:** Live manager inbox updates
- Amazon Bedrock:** AI narrative summaries & anomaly flags
- CloudWatch:** System health, latency, error rates
- AWS Lambda:** Budget rollup aggregation functions

END-TO-END LIFECYCLE (9 Steps)

- Employee logs in via Cognito
- Fills form - name auto-populated
- Uploads supporting doc to S3
- Submits - AI Agent activates
- Textract validates doc vs form
- Routed to manager dashboard
- Manager reviews with AI score
- Approve or Reject with comment
- Director sees org-wide analytics